Predicting Healthcare Insurance costs and Risk Categories Using Machine Learning

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ABSTRACT –This study develops a machine learning approach to predict insurance coverage costs and classify insured individuals into risk categories. This project uses data from 1,338 insured individuals and includes the following techniques: data preprocessing, exploratory data analysis, feature engineering, dimensionality reduction, regression modeling, and classification modeling. Insurance charges were log-transformed to reduce skewness, and engineered features, including smoking status, the BMI-smoking interaction, and age-blood pressure risk, were incorporated to improve model performance. Several models were evaluated, including linear models, support vector models, neural networks, Random Forest, and XGBoost. According to the findings, ensemble methods achieved the best results. Random Forest and XGBoost achieved the highest classification accuracy of almost 92%, while XGBoost generated the best regression result with an R^2 of 0.889. These findings suggest that very advanced machine learning techniques will be able to improve the prediction of healthcare costs and the classification of risk, due to their ability to capture complex, non-linear relationships in the data.

Index Terms – Machine Learning, Healthcare Insurance, Predictive Modelling, Classification, XGBOOST

I. INTRODUCTION

As health care costs rise in America, accurately forecasting these costs is becoming a necessity. For national health expenditures (NHE) in 2024, NHE increased 7.2 percent going from $5.3 Trillion to roughly $15,474 per capita and approximately 18 percent (%) of total GDP. Furthermore, private health insurance spending rose 8.8 percent to $1.64 trillion (~33% of total NHE). [1]

Affordability has emerged as another key public issue, along with total expenditures. Among Americans, the issue that they perceive to be the most pressing health issue in the United States is the cost of health care; a Gallup-related news article states that 29% of adults in the U.S. believe that the cost of health care is the most urgent health issue in the United States, more than access to care or obesity. [2]

In addition, rational cost prediction and risk classification are critical for insurers, health service providers, employers, and policy makers. Administrative expenses constitute a significant portion (15% to 25%) of total spending on national healthcare. [3] As a result of this administrative essential that reliable estimates of cost are produced before any expenditures can occur.

Traditionally, actuarial models have relied heavily on limited predictive parameters, such as age, sex, BMI, and smoking habits, to estimate healthcare expenditure. Though these parameters indicate significant biological risk signals, they are not comprehensive enough to capture the full spectrum of an individual's health. Recent studies using predictive analytics and machine learning (ML) techniques have shown that incorporating additional parameters can significantly improve prediction accuracy. [4]

The purpose of this research is to predict health insurance costs and classify individuals into healthcare risk groups based on their level of exposure to those risks. The main data used in this study come from a group of 1,338 insured individuals, including demographic data, lifestyle characteristics, and health information such as age, BMI, smoking status, blood pressure, how often they exercise on average, yearly income, and occupational risk. Using the regression model to predict the actual amount of insurance charges as a continuous value and the classification model to categorize the subject into one of three categories: low, medium, or high risk.

The project executes an entire machine learning workflow, including data preprocessing, exploratory data analysis, feature engineering, PCA-based dimensionality reduction, model training, cross-validation, and model evaluation. Multiple regression and classification techniques were tested, including Linear Regression, Ridge Regression, Lasso Regression, Support Vector Regression (SVR), Logistic Regression, Support Vector Machine (SVM), Multi-Layer Perceptron (MLP) Neural Network, Random Forest, and XGBoost.

The results of this study show that advanced ensemble models outperform classic statistical models in regression and classification tasks. The project demonstrates how machine learning and predictive analytics may enhance healthcare insurance risk assessment and cost prediction.

II. LITERATURE REVIEW

Machine learning is becoming increasingly significant in healthcare analytics due to its ability to process complex datasets and identify non-linear relationships between variables. The machine learning approaches can increase predictive accuracy in medical applications by leveraging large, complex, multidimensional datasets. [5] These methods have been applied extensively to patient risk assessment, healthcare cost estimation, and disease prediction.

Previous research has demonstrated that supervised machine learning approaches can estimate healthcare expenses and identify high-cost patients. Research has used models such as

Random Forests, XGBoost, and logistic regression to forecast medical expenses, demonstrating that machine learning can capture intricate risk patterns beyond conventional linear techniques. Non-linear and ensemble models can improve predictive accuracy, as interactions among demographic, behavioral, and clinical factors influence healthcare costs. [6]

Random Forest and XGBoost have gained popularity in healthcare analytics for their ability to model complex feature interactions while reducing overfitting. When comparing healthcare prediction models, studies typically find that ensemble-based and boosting methods achieve better generalization and predictive accuracy than classic linear models. [7]

Although many studies focus solely on cost prediction, few integrate regression and classification tasks within the same healthcare insurance framework. This study contributes by comparing several standard and sophisticated machine learning algorithms for continuous cost prediction and healthcare risk categorization.

III. DATASET OVERVIEW

This research's dataset consists of 1,338 records and 12 columns capturing the demographics, clinical history, lifestyle choices, and socioeconomic status of each record.

The following table illustrates the features used for this research:

TABLE I: Dataset Variables and Descriptions

Column	Type	Description
age	Integer	Age of the primary beneficiary in years
sex	Categorical	Gender of the insurance contractor (male / female)
bmi	Float	Body Mass Index (kg/m²)
children	Integer	Number of dependents covered by the plan
smoker	Categorical	Smoking status of the beneficiary (yes / no)
region	Categorical	Beneficiary's residential area
blood_pressure	Float	Resting systolic blood pressure
exercise_frequency	Categorical	Self-reported workout routine (Daily / Weekly / Rarely / Never)
pre_existing_condition	Boolean	Whether the individual had a chronic disease prior to coverage
occupation_risk	Categorical	Job-related physical hazard level (Low / Moderate / High)
annual_income	Float	Estimated yearly earnings of the beneficiary (USD)
charges (target)	Float	Individual medical costs billed by health insurance

The target variable, charges, is the actual cost billed by the insurance plan for each beneficiary. It is a continuous variable used as a prediction target in a regression model and as a label in a classification model.

IV. METHODOLOGY

A. Data Preprocessing

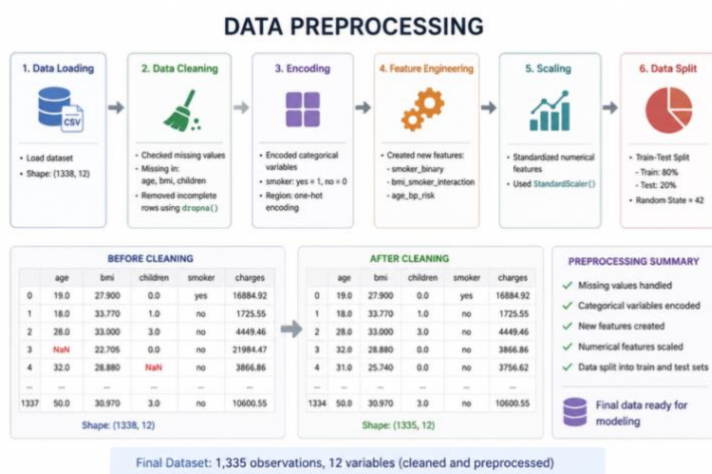

Figure 1: Data Processing and Cleaning

Data cleaning, missing value treatment, feature encoding, feature scaling, and train/test splitting are all steps in preprocessing the dataset before modeling.

The dropna() function was used to remove Missing Values from the dataset, ensuring the data remained consistent. StandardScaler was used to standardize numerical features, and OneHotEncoder was used to encode categorical variables. A preprocessing pipeline was created and executed using the Pipeline and ColumnTransformer methods in Scikit-learn.

The dataset was split into training, validation, and test sets using a stratified split, ensuring each set maintained the class balance of the entire dataset across all risk classes.

The preprocessing framework ensured that every model input was the same numeric type and prepared for use in machine learning algorithms. Scaling is important for distance-based models such as SVR and SVM. In contrast, categorical variables such as sex, whether they were smokers, the part of the country they lived in, how often they exercised, and their occupation-based vulnerability to becoming ill were encoded in the models using one-hot encoding. Since all models were utilized using the same preprocessing pipeline, the comparisons were equal and therefore reasonable.

V. EXPLORATORY DATA ANALYSIS (EDA)

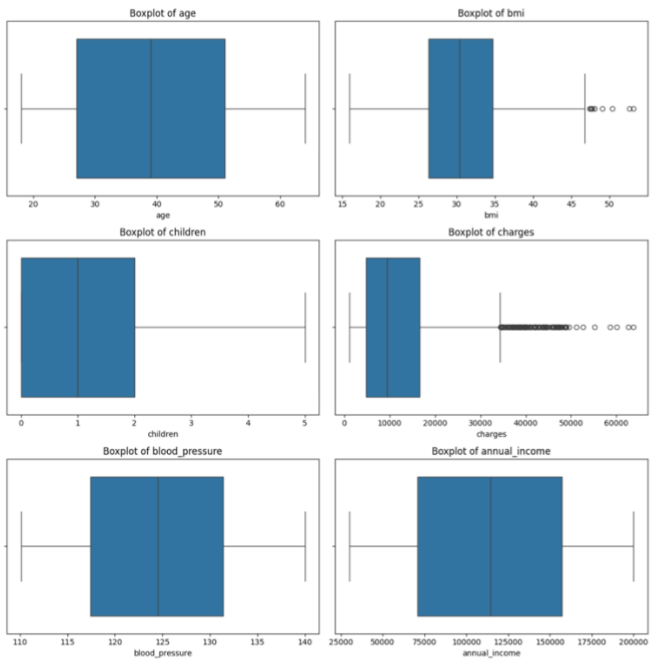

Figure 2: Boxplot distributions of numerical variables

The Exploratory Data Analysis (EDA) was conducted to understand the dataset's structure, distribution, and overall quality before developing models. The final, cleaned dataset had 1,335 observations and an average age of ~39 years, with a similarly high average BMI of ~30.7, meaning that many individuals, on average, fall within the overweight category. This data showed high variability in insurance charges, with the highest amount at ~$63,000. In addition, categorical variables provided evidence that most individuals in this study were non-smokers, most frequently lived in the Southeastern United States, and worked in a moderate-risk occupation. Examination of histograms indicated strong right skewness in insurance charges, with the majority of individuals incurring either low or moderate insurance costs, while only a small number incurred very high costs.

The presence of several outliers associated with insurance charges and various BMI-related variables was corroborated by boxplots, whereas age, blood pressure, and annual income showed much less variability. Overall, the exploratory data analysis revealed high skewness and significant outliers in insurance charges, suggesting the need to apply a log transformation and robust pre-processing before training the model.

VI. FEATURE RELATIONSHIP

The analysis of feature relationships was carried out using a correlation heat map to examine the relationships between numerical variables and insurance charges. The figure 3 showed a correlation of 0.89 between log charges and charges, indicating that transforming the original cost pattern using logarithmic transformation reduces some of the skewness in the original cost distribution. Furthermore, smoking-related variables, including smoker binary and BMI-smoker interaction, were positively correlated with insurance charges and indicate that smoking behavior is a significant factor in determining costs. Conversely, there was very little correlation between annual income and insurance charges (~0.00), and the correlation between blood pressure and insurance charges was very weak (-0.04). Overall, the results of this analysis show that smoking behavior (or smoking status) and age-related risk are stronger determinants of insurance costs compared with income.

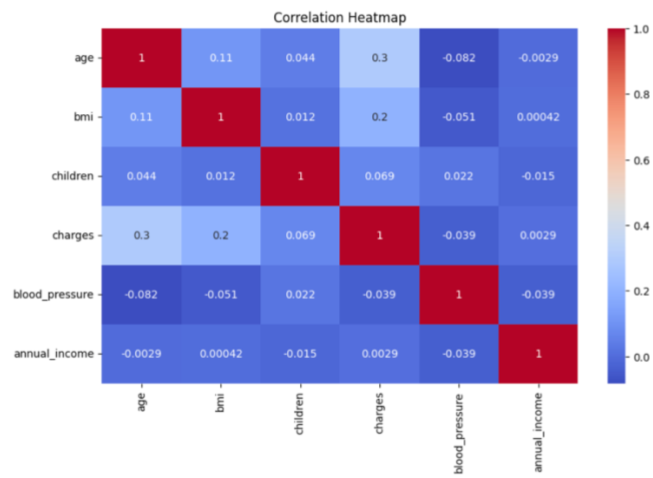

Figure 3: Correlation heatmap

VII. HANDLING SKEWNESS: LOG TRANSFORMATION

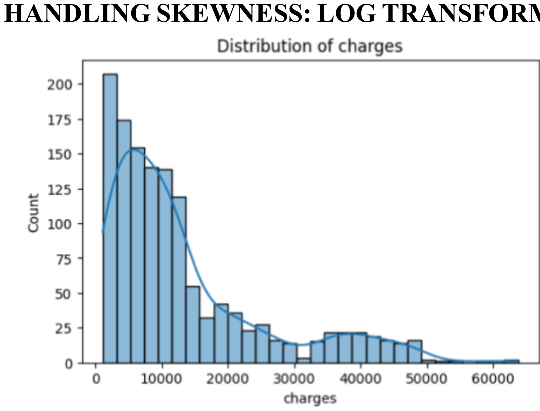

Figure 4: Distribution of medical charges before Log Transformation

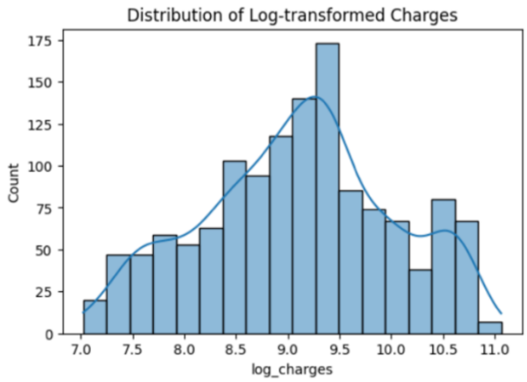

Figure 5: Distribution of medical charges after Log Transformation

The data on insurance costs are predominantly right-skewed, as reflected in a cost range of approximately $1,121.00 to $63,770.00 and an average of $13,255.00 (the mean insurance cost), compared to the median cost of $9,378.00, suggesting that high-value extreme observations inflated the average insurance cost shown in figure 4. High levels of skewness can negatively impact the regression model by overweighing a small number of extremely high observations.

The charges variable was log-transformed to resolve this problem, yielding the new target variable log_charges shown in figure 5. After this transformation, the range of values for the new target variable was approximately 7.0 to 11.0, and the distribution was very close to normal, with greater symmetry than before the transformation. The result of this transformation was a reduction in the influence of extreme outliers and improved stability in the regression modeling process. In summary, a log transformation enabled the models to identify more clearly defined patterns in healthcare costs without being overwhelmed by very high insurance charges.

VIII. FEATURE ENGINEERING AND MODELLING SETUP

Feature engineering techniques to develop variables that better reflect risk patterns in healthcare. Smoking status was transformed into smoker_binary (with 1 meaning "a smoker" and 0 meaning "a non-smoker"), whereas bmi_smoker_interaction was measured to enable the combined effect of BMI and smoking on risk. Age and blood pressure were used to generate a new feature, age_bp_risk, that shows their impact on risk.

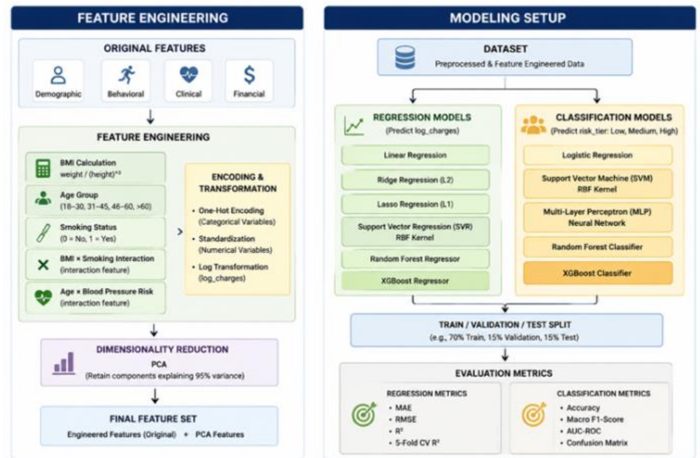

Figure 6: Feature Engineering and Modeling Setup

Each charge assigns an insurance charge to one of the three quantile-based risk tiers. There were approximately 445 observations in each of the three quantile-based risk tiers: Low Risk, Medium Risk, and High Risk. The log-charges served as the regression target, and each of the three risk tiers served as the classification target. The dataset is divided into 70% for training, 15% for validation, and 15% for testing shown in figure 6. The numeric variables were both scaled, while the categorical variables underwent one-hot encoding.

IX. PRINCIPAL COMPONENT ANALYSIS (PCA)

The Principal Component Analysis (PCA) was used to minimize the dimensionality of the feature matrix while retaining as much data as possible from the original data set. Figure 7 shows the first five components or variables accounted for roughly 88% of the variance in the original data set. Approximately six variables accounted for 95% of the variance in the original data set. PCA algorithms process the features, reducing the original 25-dimensional feature space to a new 14-dimensional one.

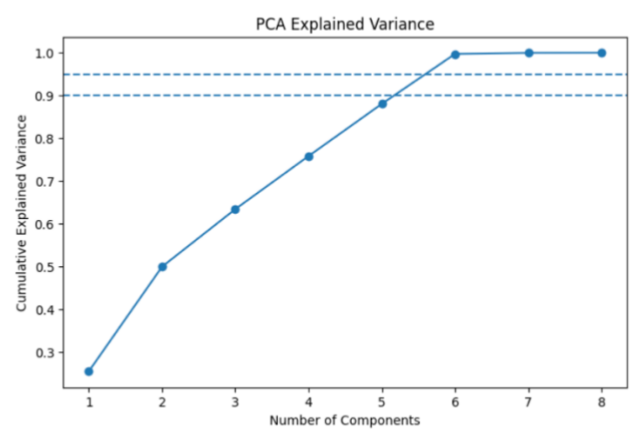

Figure 7: Principal Component Analysis

In terms of model performance using the two feature sets, the R^2 value for the original feature set was 0.855, and the R^2

value for the PCA-reduced feature set was 0.841, indicating that PCA reduced model complexity and resulted in a loss of model performance (consistent with general findings). However, in this case, the original feature set performed slightly better than the PCA-reduced feature set.

X. MODEL DEVELOPMENT

After completing preprocessing and feature engineering, we trained two sets of models. In the regression task, we trained models (linear regression, Ridge regression, Lasso regression, Support Vector Regression, Random Forest, and XGBoost) on log charges as the target variable to predict insurance costs. The purpose of training regression models is to compare simple linear regression, regularized models, non-linear methods, and ensemble regression.

For classification tasks, we trained models (logistic regression, support vector machine, multi-layer perceptron neural network, random forest, and XGBoost) using risk tier as the target variable. The purpose of training these classification models is to compare baseline classifiers (logistic regression), non-linear classifiers (support vector machines), neural network classifiers (multi-layer perceptron), and ensemble classifiers (random forests). All models were evaluated using task-specific metrics in the results section.

XI. EVALUATION METRICS

The model uses separate metrics for regression and classification, ensuring that evaluations for each task type are appropriate. On regression tasks, MAE, RMSE, and R^2 were used to assess overall error and model explanatory power. MAE measures how large the errors are on average when predicting the value of an observation, while RMSE assigns a penalty based on how large any given error margin is. R^2 indicates the proportion of variation in insurance costs that the model explains.

For the classification task, Accuracy, Macro F1-score, AUC-ROC, and confusion matrices were used to evaluate overall correctness, performance balance across risk tiers, ability to separate classes, and specific misclassification patterns for each class. All these metrics provide a comprehensive evaluation of both predictive accuracy and the model's reliability.

XII. RESULTS AND DISCUSSIONS

A. *Regression Model performance*

TABLE II: Regression Model Performance Comparison

Model	MAE	RMSE	R2
Linear	0.2206	0.3553	0.8418
Ridge	0.2205	0.3551	0.8419
Lasso	0.2411	0.3785	0.8204
SVR	0.2163	0.3402	0.855

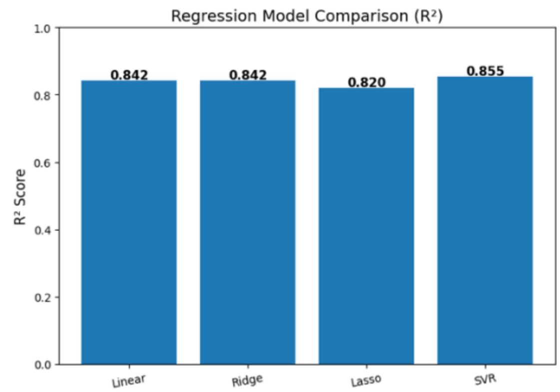

Figure 8: Regression Model Comparison based on R^2 Score

Linear regression, ridge regression, lasso regression, and support vector regression (SVR) evaluate for the target variable log_charges. With an R^2 of 0.855, SVR had the best test performance, demonstrating its capacity to capture non-linear correlations in healthcare insurance costs. Model performance is evaluated using MAE, RMSE, R^2, and 5-fold cross-validation. Strong model stability and generalization were shown by the highest mean cross-validation R^2 values of roughly 0.770 obtained from Ridge Regression and Linear Regression shown in table II and figure 8. Lasso Regression, on the other hand, performed the worst overall. These findings imply that while Ridge and Linear Regression provide consistent baseline performance, non-linear techniques increase predicted accuracy.

B. *Classification Model Performance*

TABLE III: Classification Model Performance Comparison

Model	Accuracy	Macro F1	AUC-ROC
Logistic Regression	0.8358	0.8354	0.9397
SVM	0.8706	0.8709	0.9558
MLP Neural Network	0.8458	0.8448	0.9427

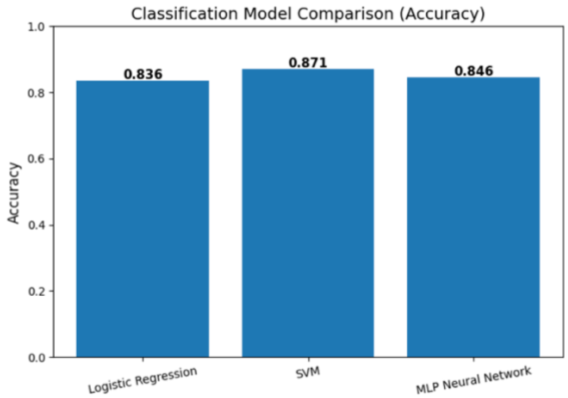

Figure 9: Classification Model Comparison based on Accuracy

For healthcare risk classification, logistic regression, support vector machines (SVMs), and MLP (Multi-Layer Perceptron) Neural Network were evaluated using accuracy, Macro F1-score, AUC-ROC, and confusion matrices to assess the models. The table III and figure 9 shows that SVM outperformed the other models, achieving an overall accuracy of over 87% and demonstrating strong classification performance across the Low, Medium, and High-risk categories.

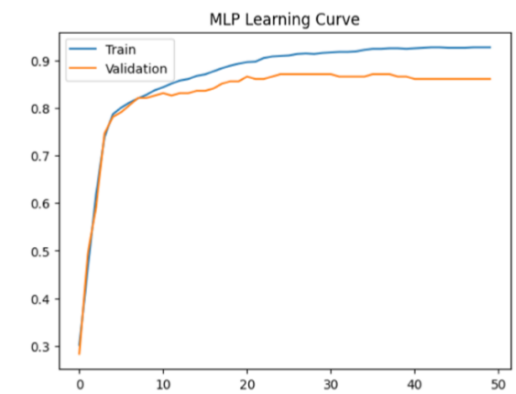

Figure 10: MLP neural network learning curve for training and validation performance.

During training, the MLP (Multi-Layer Perceptron) Neural Network demonstrated robust learning behavior, with training accuracy rising to roughly 93% and validation accuracy stabilizing at 86-87%. The relatively minor difference in training and validation performance suggests slight, but not severe, overfitting. Logistic regression showed stable baseline performance but lower total accuracy than SVM and MLP. Overall, the results show that SVM was the most effective non-linear classification model for this dataset.

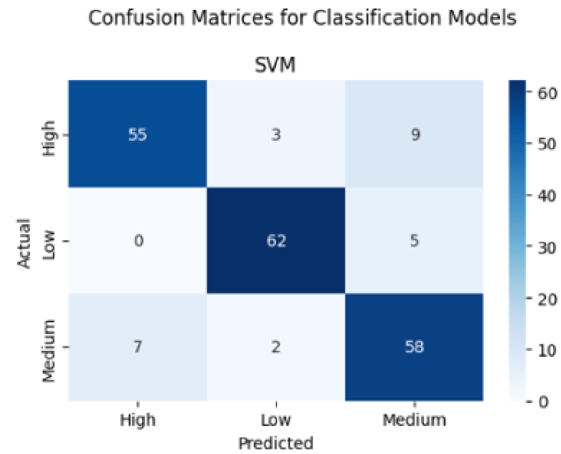

Figure 11: Confusion matrix of the SVM classification model

Figure 11 shows the evaluation of the confusion matrices revealed that the SVM algorithm showed balanced performance across all 3 types of classification (high risk, medium risk, and low risk) and correctly classified 55 high, 62 low, and 58 medium-risk observations as either being high, medium, or low risk. There were only 9 observations that were incorrectly predicted to be high risk when they were medium risk, and 5 that were incorrectly predicted to be low risk when they were actually medium risk. It indicates that the SVM classifier performed well at discriminating among the three risk categories within the healthcare system and provided an accurate and reliable measure of performance across all three categories.

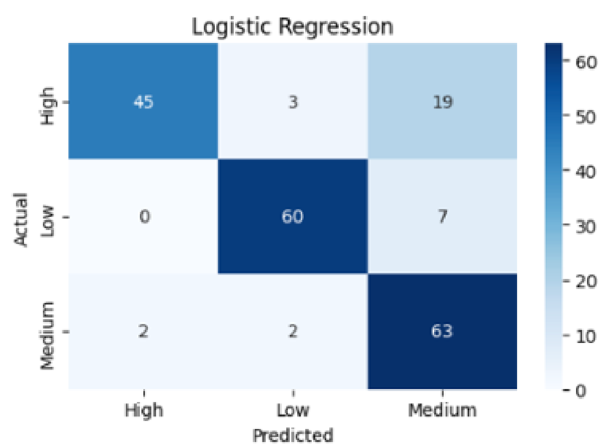

Figure 12: Confusion matrix of the Logistic Regression

Logistic regression confusion matrix indicating strong performance in accurately identifying low-risk individuals based on 60 low-risk observations shown on figure12. In addition to correctly predicting 63 medium risk cases and 45 high risk cases, the greatest classification error was 19 high risk individuals incorrectly classified as medium risk, suggesting a significant degree of overlap between these two risk categories. Overall, logistic regression's results indicate it is a solid foundation for classification; however, it doesn't have the ability to separate the higher-risk groups.

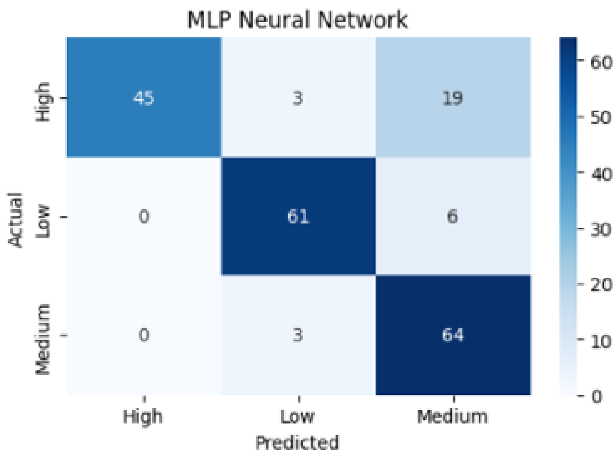

Figure 13: Confusion matrix of the MLP Neural Network

The MLP Neural Network's Confusion Matrix in figure 13 showed promising classification results, especially for the medium-risk group, which had 64 of its observations correctly classified. The model also predicted correctly 61 low-risk cases and 45 high-risk cases. As in Logistic Regression, the major classification error was misclassifying 19 individuals in the high-risk group as medium risk. Even though there is some overlap between the high- and medium-risk groups, the MLP Neural Network achieved good overall classification accuracy and good learning behavior on the data.

Overall, the confusion matrices show that both the High Risk and Medium Risk groups share similar characteristics, making it difficult to distinguish between them. The conclusions drawn from the analysis of the confusion matrices indicate that SVM provides the highest classification quality compared with the other non-linear classifiers tested.

C. Ensemble Model Comparison

In addition to exploring traditional regression and classification models, we also explored ensemble methods such as Random Forests and Gradient Boosting. Ensemble models combine the predictions of multiple decision trees to increase prediction accuracy, reduce overfitting, and improve the ability to capture complex nonlinear relationships in healthcare datasets, compared to traditional methodologies. Random Forest uses a technique called bagging, which provides stability and reduces the variance of each prediction, while XGBoost uses sequential boosting, which optimizes overall predictive performance and minimizes errors in the model. We completed our analyses of ensemble learning capabilities by evaluating the performance of ensemble models in regression and classification tasks for predicting costs associated with health insurance and classifying risk for a healthcare organization.

Table IV: Advanced Regression Model Performance Comparison

Model	MAE	RMSE	R2
Random Forest	0.1645	0.3033	0.8847
XGBoost	0.1586	0.2971	0.8894

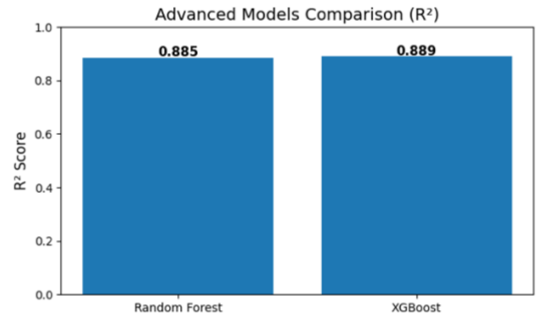

Figure 14: Advanced Regression model comparison based on R^2 score.

In regression, XGBoost outperformed the others overall, with an R^2 of 0.889 and an MAE of around 0.159. Random Forest also performed well, demonstrating that both methods achieve a good level of accuracy, with $R^2 = 0.885$ and MAE = 0.165 shown in table IV. In comparison, XGBoost performed slightly better than Random Forest shown in the figure 14.

Table V: Advanced Classification Model Performance Comparison

Model	Accuracy	Macro F1	AUC-ROC
Random Forest	0.9204	0.9202	0.965
XGBoost	0.9204	0.9198	0.9635

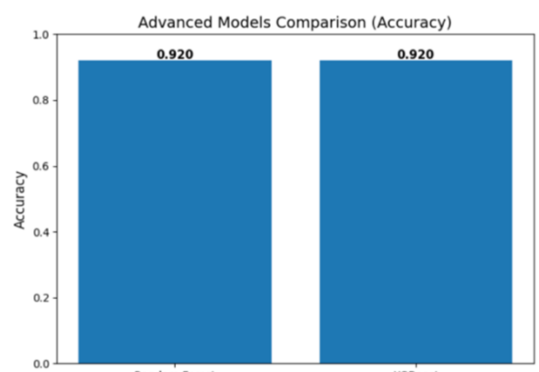

Figure 15: Advanced Classification Model Performance Comparison

For Classification, Random Forests and XGBoost achieved around the same 92% accuracy as shown in table V. Both models were able to make almost equally balanced and steady predictions in the three risk categories, Low, Medium, and High, as shown by their Macro F1 Scores shown in figure 15. Traditional regression and classification techniques were less effective than ensemble approaches, and XGBoost emerged as the top-performing regression technique in this study.

XIII. CONCLUSION

The project goal was to use machine learning algorithms to build predictive models of healthcare insurance costs and to classify individuals into healthcare risk categories based on a set of variables: demographic, behavioral, clinical, and financial. The methods used for the analysis in this project included data preprocessing, exploratory data analysis, feature engineering, principal component analysis (PCA), and regression and classification analyses.

The research demonstrates that ensemble methods achieve statistically superior performance compared to traditional statistical models. XGBoost achieved the highest R^2 of 0.889. Random Forest and XGBoost achieved the best classification results, with accuracies approaching 92%. The analysis indicates that health insurance costs exhibit a complex, non-linear relationship that is better represented by advanced machine learning techniques.

The study showed that feature engineering produced significant gains in model performance as compared to increasing model complexity. For example, smoking status, the interaction between age and BMI with the risk of hypertension due to high blood pressure or both, as well as the interaction between age and being overweight, were important in determining the patterns of behaviors related to health and lifestyle. Although PCA reduced dimensionality at the expense of only a small loss in performance, the original feature set achieved slightly better overall performance.

The research indicates that various Machine Learning (ML) methods can create effective approaches for assessing Healthcare Insurance Risk, with an emphasis on Ensemble Modeling. The project identifies a need for both a multidimensional Healthcare Dataset and appropriate data preprocessing expectations to yield greater accuracy and data-driven value in the decision-making style used today in Insurance Analytics through an enhanced method of Advanced Modeling.

XIV. FUTURE WORK

In the future, the study could expand by using machine learning and deep learning techniques to predict health insurance claims. Some examples of these techniques include Long Short-Term Memory (LSTMs), deep neural networks, and transformer-based methods. Utilizing an ensemble approach to combine predictions, as well as automated hyperparameter tuning, will likely help create better predictive models and improve their stability.

The study could further use larger databases of real-world healthcare claims data (e.g., longitudinal patient records, hospital admissions) to develop and validate a robust model. The use of explainable AI techniques (like SHAP) will improve the interpretability of the model's results. It will help identify the most significant contributors to a person's risk of health-related events and provide external validation of the model to demonstrate that it produces reliable and generalizable results across multiple populations and settings.

XV. REFERENCES

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